

Nathalie Rodriguez Mendoza

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EDUCATION

Rice University

Houston, TX

Bachelor of Science in Computer Science — GPA: 3.92/4.00

Expected Graduation: May 2029

Relevant Coursework: Multivariable Calculus, Discrete Math, Linear Algebra, Probability and Statistics, Data Structures and Algorithms

TECHNICAL SKILLS

Programming Languages: Python, Java, HTML/CSS, TypeScript, JavaScript

Frameworks: React, Next.js, Figma, Git

Certifications: MTA Introduction to Programming: **Python** and **Java**, IT Specialist: **HTML/CSS**, **Networking**, and **Data Analytics**, IBM: **Artificial Intelligence Fundamentals**

Languages: English (Proficient), Spanish (Proficient)

EXPERIENCE

Algorithms Instructor

May 2026 – Present

Tapia Center, Rice University

Houston, TX

- Enabled **720 campers** to build admissions algorithms selecting 200 of 1,000 applicants by teaching Boolean logic
- Coordinated daily transitions for 180 campers across counselors and instructors, maintaining **100%** on-time session starts throughout the 4-week program
- Led the program's largest presentation cohort by mentoring **40 students**, with one team earning Best Overall

Undergraduate Research Assistant

January 2026 – May 2026

Rice University

Houston, TX

- Built and evaluated Random Forest models on **20,000** material samples to predict Curie temperatures using **Python** and **scikit-learn**
- Reduced prediction RMSE **40%** from 50K to 30K through data cleaning and hyperparameter tuning

Software Developer

January 2026 – May 2026

Fondren Library, Digital Media Commons

Houston, TX

- Digitized **60** paper forms by building a full-stack library administration system with **React** and **Supabase**
- Reduced check-in time by **50%**, from 10 to 5 minutes, by automating form submission and record storage
- Eliminated paper-based future entries by integrating React forms with **Supabase APIs** for centralized data retrieval

PROJECTS

Feeding South Florida Access Map | *React, MapLibre GL, FastAPI, PostgreSQL, Census API*

- Shipped a full-stack mapping tool adopted by food-bank staff to target outreach across **1,497 tracts** in 3 counties
- Engineered a Python pipeline scoring tracts on 7 Census/USDA need indicators via percentile-rank normalization
- Implemented a **geospatial gap-analysis** algorithm to surface high-need, uncovered neighborhoods

MediMetrics | *React, Next.js, Firebase, OpenCV, MediaPipe*

- Developed a computer vision application using **OpenCV** and **MediaPipe** to extract accurate limb measurements from user-uploaded images for remote prosthetic fitting
- Built a **React** frontend with **Firebase Auth** and image upload to visualize measurement results
- Led a team of **5** by organizing standups and delegating tasks, ensuring timely milestone completion

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Hanszen College

May 2026 – Present

Website Manager

- Delegate development tasks and manage project timelines to ensure consistent site updates
- Increased website traffic by **30%** by optimizing content, improving UX, and coordinating regular updates

Other Activities and Societies: RiceApps Head of Social Media, Hanszen College Website Manager, Society of Women Engineers, CSters, HeadStart Fellowship, SEO First Year Academy